

Daily Review

Booklet p.8-9 — Space Exploration and the ISS

- 1 Name two risks of crewed space exploration.
- 2 Why does a craft far from the Sun need more cells?
- 3 What energy transfer happens in a solar cell?
- 4 What is a pressure differential?
- 5 What protects a craft on re-entry?

Fuel, radiation,
pressure, re-entry

The light is more spread
out

Light to electrical
energy

Air inside, vacuum
outside

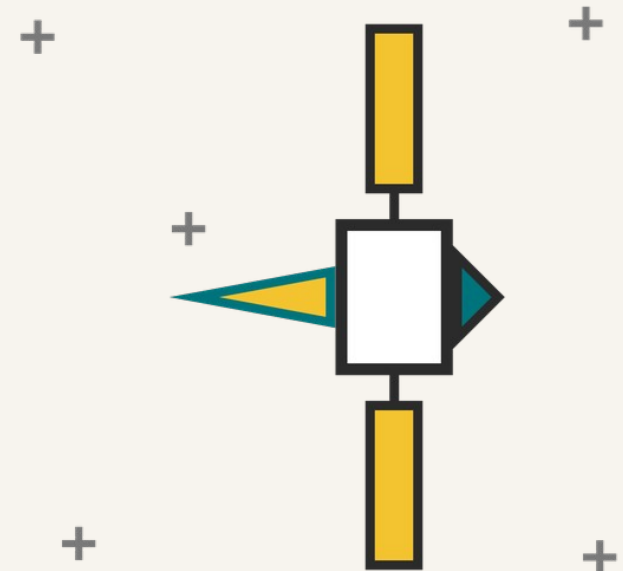
The heat shield

SPACE

Getting Around in Space

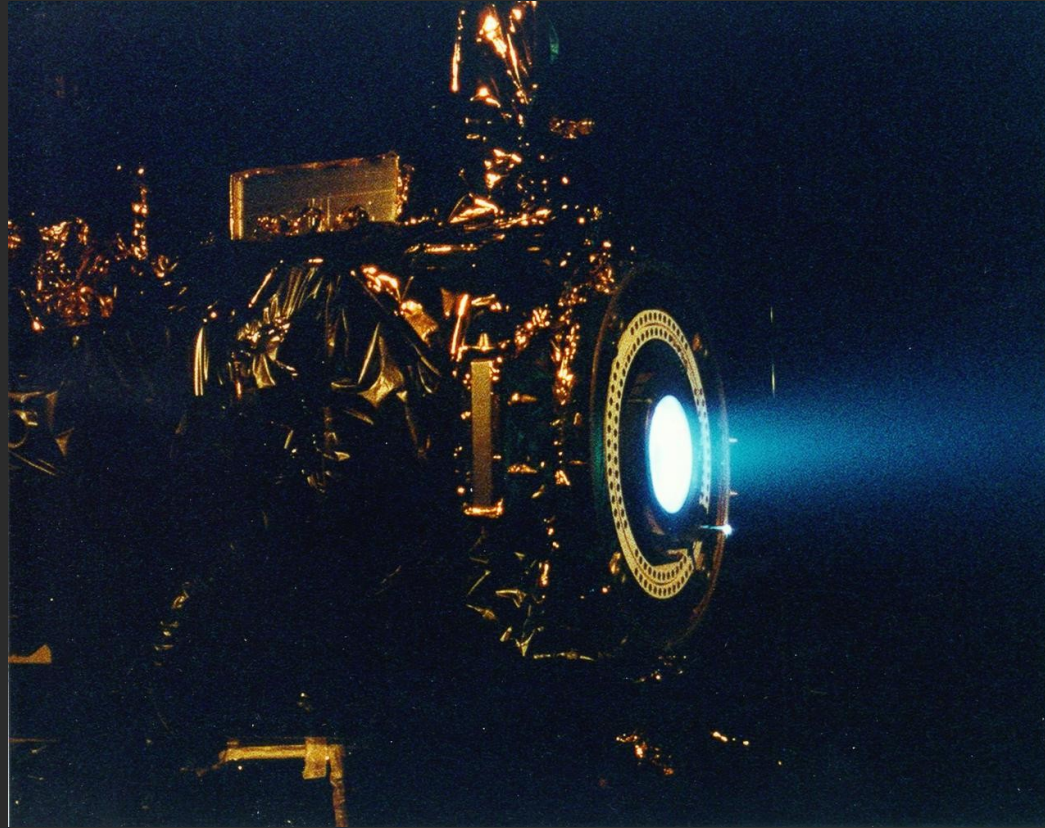
No air, no friction, no brakes.

National 5 Physics (S3)



THE BIG QUESTION

This engine's push is about the weight of a sheet of paper.
How can it be the fastest way to the outer Solar System?



The Deep Space 1 ion engine on test — NASA/JPL

PART 1

Moving and steering

How do you speed up, slow down and steer with nothing to push on?

By the end of this part...

- 1 I can describe the difficulties of manoeuvring in deep space
- 2 I can explain how a spacecraft changes speed and direction

Show me — Describe each word

SHOW ME BOARDS

1 friction

A force that opposes motion

2 thrust

The push from an engine

3 balanced forces

Equal forces in opposite directions

4 unbalanced force

A force that changes the motion

5 deep space

Far from any planet or star

Show me — Describe each word

SHOW ME BOARDS

6 manoeuvre

A planned change of motion

7 docking

Joining two spacecraft together

8 ion drive

An engine with a tiny, long thrust

9 catapult

Using a planet's gravity to speed up

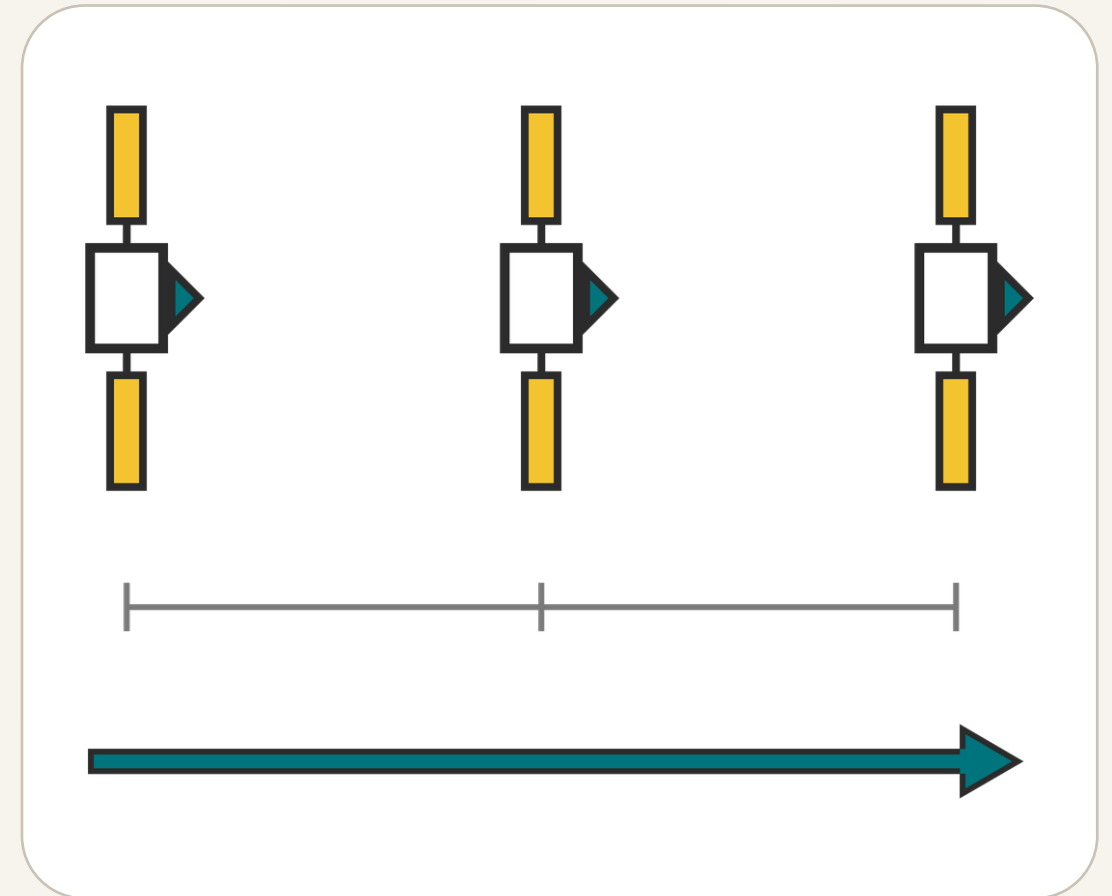
10 fuel

The store of energy a rocket burns

No friction in deep space

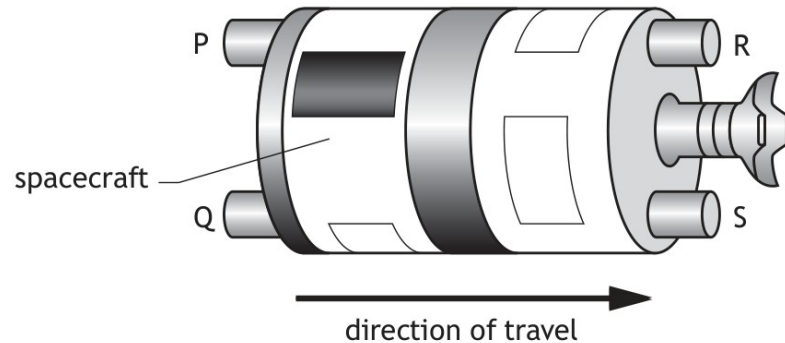
Booklet p.10

- Far out in space there is no air, so no friction.
- Engine off: the forces on the craft are balanced.
- It keeps moving at a constant speed in a straight line.



Show me – which engines?

SHOW ME BOARDS



1

Which two engines slow the craft down?

R and S

2

Which two engines speed it up?

P and Q

3

All engines off.
Describe its motion.

Constant speed

4

Why does it not slow down on its own?

No friction

Docking with the ISS

Booklet p.10, Tasks 1 and 2



- Both orbit at 7.7 km s^{-1} .
- Only the difference in their speeds matters.
- Tiny thruster pushes close the last few metres.
- With no friction, every push must be cancelled.

Show me – moving in space

SHOW ME BOARDS

EXTRA PRACTICE

- 1 A probe switches its engine off in deep space. What happens?
- 2 Why is there no friction in deep space?
- 3 How does a spacecraft slow down?
- 4 A craft starts to spin. How can it stop spinning?
- 5 Why is docking so difficult?

It keeps going, same speed

There is no air

It fires a thruster forwards

Fire a thruster the other way

No friction, and both move fast

Tasks 1 and 2 – dock with the ISS

Booklet p.10

Try to dock with the ISS in the simulator.
Then describe why it was difficult.

Booklet p.10, Tasks 1 and 2



[Docking simulator](#)

Task 2 – Answers

ANSWERS

Booklet p.10

- There is no friction, so once it moves, it keeps moving.
- Every push must be cancelled by a push the other way.
- Small thrusters give tiny changes, so it takes a long time.
- The ISS and your craft are both moving very fast in orbit.

Daily Review

Booklet p.10 — Part 1

- 1 Why is there no friction in deep space?
There is no air
- 2 Engine off in deep space: what happens?
It keeps going at a steady speed
- 3 How does a spacecraft slow down?
It fires a thruster forwards
- 4 What makes docking with the ISS hard?
No friction; both move fast
- 5 What is thrust?
The push from an engine

PART 2

Getting there faster

Two ways to cross the Solar System — and neither is brute force.

By the end of this part...

- 1 I can describe the purpose and function of the catapult
- 2 I can describe the purpose and function of an ion drive

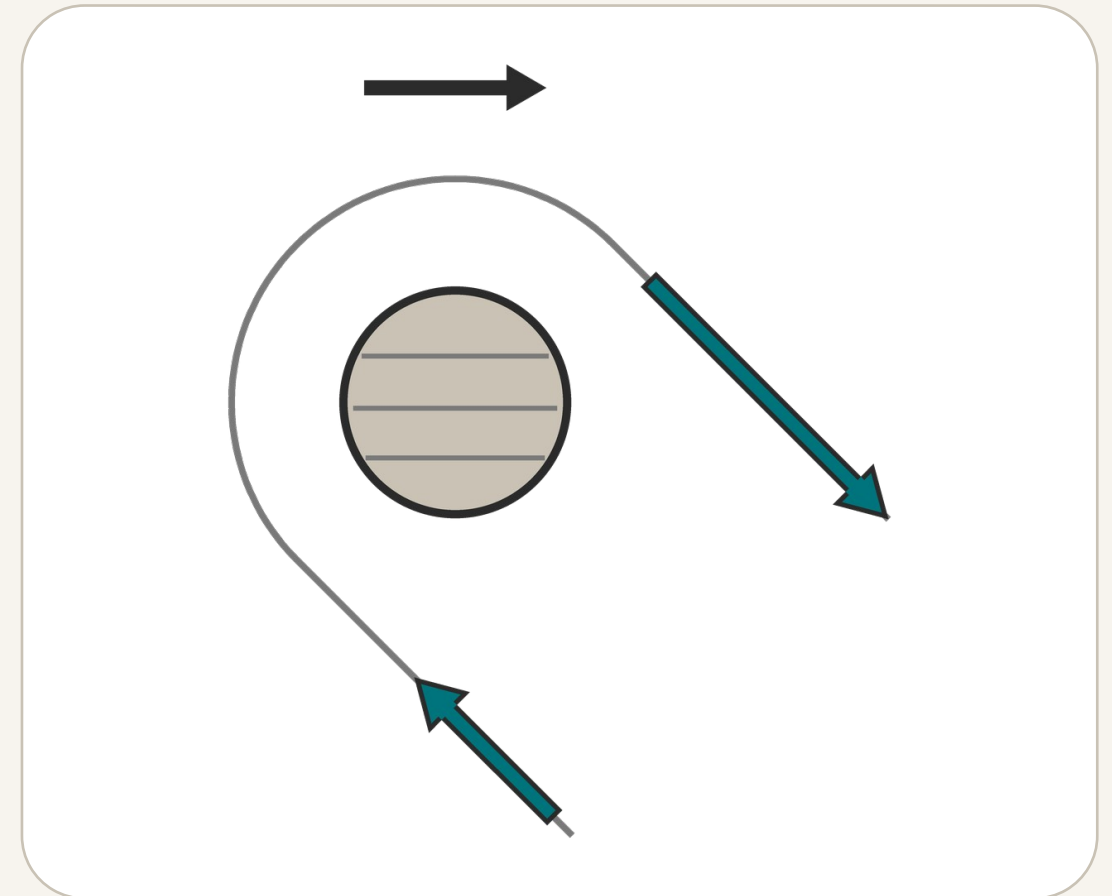


Jupiter — NASA Science / Voyager and Cassini missions

The catapult

Booklet p.10, Task 3

- A craft is steered close to a moving planet or moon.
- Its gravity pulls the craft round and speeds it up.
- It gains speed without burning fuel.
- Artemis II used the Moon's gravity to curve its path home.



Task 3 – the catapult

Booklet p.10

Watch the clip about the gravity assist.
Then describe it in your own words.

Booklet p.10, Task 3

Show me – the catapult

SHOW ME BOARDS

- 1 What is another name for the catapult manoeuvre?
- 2 What pulls the craft towards the planet?
- 3 What happens to the craft's speed?
- 4 Why does this save fuel?
- 5 Name an object a craft could catapult from.

A gravity assist

The planet's gravity

It increases

No engine burn is
needed

A planet, moon or
asteroid

Show me – the catapult

SHOW ME BOARDS

EXTRA PRACTICE

- 1 Does the catapult need the engine to be on?
- 2 What changes as well as the craft's speed?
- 3 Why does a mission to Neptune use a catapult?
- 4 Why must the craft be steered very accurately?
- 5 What if it passed too close?

No

Its direction

To save fuel and time

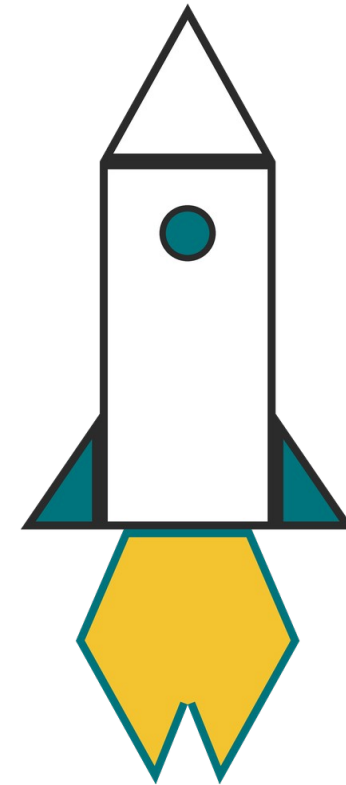
It must pass at the right distance

It could be pulled in

Chemical rockets

Booklet p.10, Task 4

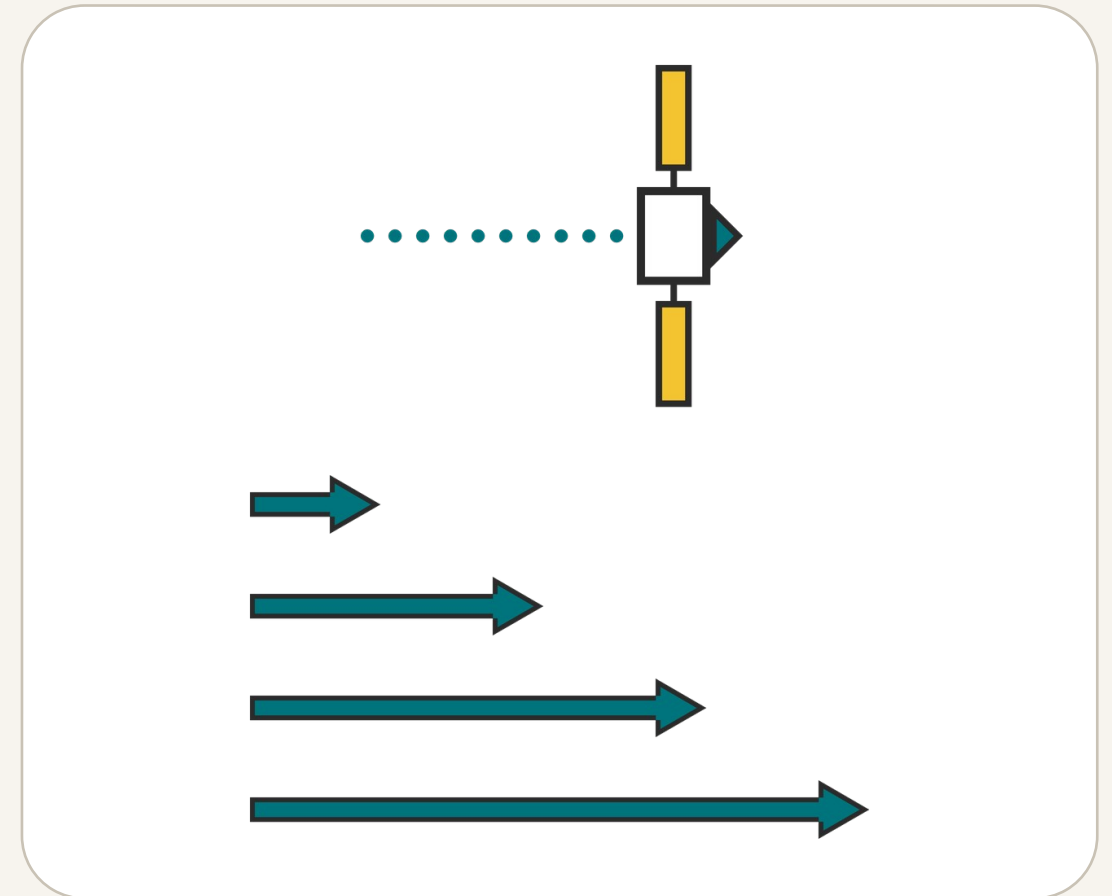
- Burning fuel pushes hot gas out very fast.
- A huge thrust, but only for a few minutes.
- Lift-off needs a thrust bigger than the weight.



The ion drive

Booklet p.10, Task 4

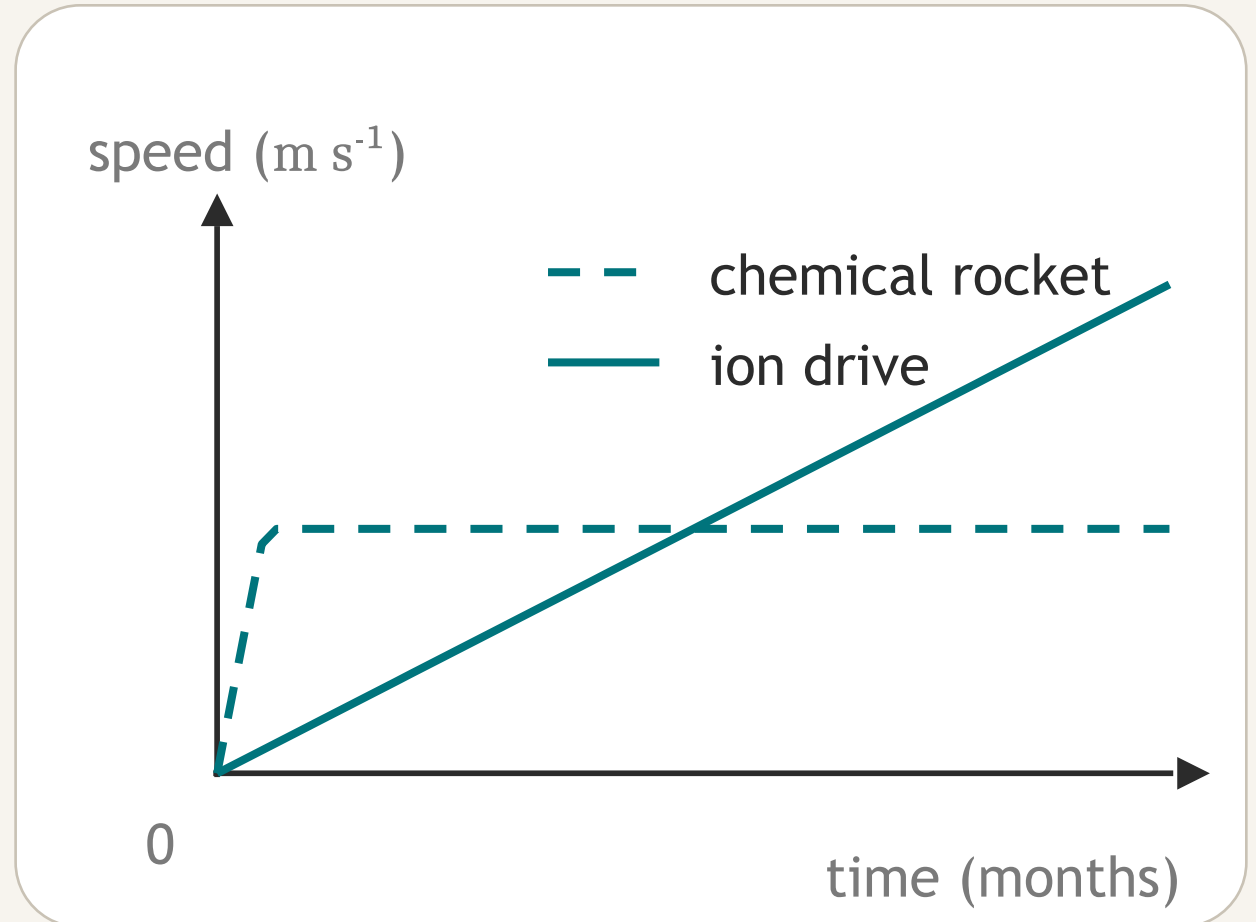
- Pushes out charged particles, using very little fuel.
- A small unbalanced force, over a very long time.
- So the craft reaches a very high speed in the end.



Why the ion drive wins in the end

Same journey, two engines

- The ion drive's force is tiny, so a is tiny.
- But it fires for months, so the speed keeps climbing.
- The rocket burns out in minutes, then coasts.
- Given time, the ion drive ends up far faster.



Agree, disagree or it depends?

SHOW ME BOARDS

Boards up: AGREE, DISAGREE or IT DEPENDS — and why

- 1 A spacecraft needs its engine on to keep moving.
Disagree: no friction out there
- 2 There is no gravity in deep space.
Disagree: it is just very weak
- 3 An ion drive is more powerful than a rocket.
Disagree: its thrust is tiny
- 4 The catapult speeds a craft up without fuel.
Agree: gravity does the work
- 5 You cannot steer without air.
Disagree: fire a thruster

Hinge question

FINGER VOTE

Think first — then show A to E on your fingers, 1 to 5

An ion drive's thrust is only about 0.1 N.
The main reason it still reaches a very high speed is

- A it has no mass to accelerate
- B it keeps pushing for months on end
- C its thrust grows as it travels
- D there is no gravity in deep space
- E it carries more fuel than a rocket.

Answer B

Show me – engines

SHOW ME BOARDS

EXTRA PRACTICE

- 1 What must be true of the thrust at lift-off?
- 2 Why are ion drives used for long journeys?
- 3 Why is a small force useful in deep space?
- 4 Give one drawback of an ion drive.
- 5 Which engine uses fuel faster?

Bigger than the weight

A tiny force, kept up for years

Nothing slows the craft down

It cannot lift off

The chemical rocket

Task 4 – types of rocket engine

Booklet p.10

Fill in the table for a chemical rocket
and for an ion drive.

Booklet p.10, Task 4

Task 4 – Answers

ANSWERS

Booklet p.10

Engine	Description	Benefits	Drawbacks
Chemical	burns fuel; hot gas pushed out	huge thrust; can lift off	uses fuel very fast
Ion drive	pushes out charged particles	very little fuel; high top speed	tiny thrust; cannot lift off

Would you rather...

TALK IT OUT

...send a probe to Jupiter on an ion drive,
...or catapult it past Venus and Earth first?

Which gets there sooner? Which uses less fuel?
What could go wrong with each?

Extension — space in everyday life

Booklet p.11

Find out how space research has
changed things we use every day.

Booklet p.11, Extension



[NASA at home](#)

Checkpoint

- 1 Why does a craft in deep space keep moving with its engine off?
No friction: the forces are balanced
- 2 How can an ion drive's tiny force give a very high speed?
It acts for months on end
- 3 Describe the catapult manoeuvre, and say why it is used.
Gravity speeds it up; it saves fuel

Exam practice

Past paper

- National 5 Physics 2025, Section 2, question 4(a). 2 marks.
- Question 4(b)(i). 1 mark. A probe on its way to an asteroid.
- Then booklet p.11, Task 5.

LOOKING AHEAD

Satellites

What do satellites do for us — and why does height set the orbit?

Plenary

- 1 The big question: how can a tiny push be the fastest way out there?
- 2 Why is there no friction in deep space?
- 3 Engine off, deep space: describe the motion.
- 4 Describe how an ion drive works.
- 5 Describe the catapult manoeuvre.

It never stops pushing

There is no air

Constant speed, straight line

A tiny thrust, for a very long time

Gravity pulls it round, faster

Plenary

EXTRA PRACTICE

- 1 How does a spacecraft change direction?
- 2 What makes docking with the ISS difficult?
- 3 Give one benefit of an ion drive.
- 4 What does the catapult change about the motion?
- 5 Which engine gives the bigger thrust?

It fires a thruster

No friction; both move fast

It uses very little fuel

Its speed and its direction

The chemical rocket

Extension

- 1 Explain why a probe's engine can be off for years of its journey.
- 2 Suggest why a craft uses both a chemical rocket and an ion drive.
- 3 Suggest why a catapult cannot be used on every part of a journey.

Extension

EXTRA PRACTICE

- 1 A thruster fires for two seconds to start a spin. How is it stopped?
- 2 Why would an ion drive be useless in Earth's atmosphere?
- 3 Suggest why a probe to Saturn might pass Venus first.

Course specification – 1 of 3

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the challenges of space travel:
 - travelling large distances with the possible solution of attaining high velocity by using ion drive (producing a small unbalanced force over an extended period of time)
 - travelling large distances using a ‘catapult’ from a fast moving asteroid, moon or planet

Course specification – 2 of 3

N5 Physics course specification v4.0, p.8 – Space exploration

- Awareness of the challenges of space travel:
 - manoeuvring a spacecraft in a zero friction environment, possibly to dock with the ISS
 - maintaining sufficient energy to operate life support systems in a spacecraft, with the possible solution of using solar cells with area that varies with distance from the Sun

Energy for life support was taught last lesson.

Course specification – 3 of 3

N5 Physics course specification v4.0, p.6 – Newton's laws

- Application of Newton's laws and balanced forces to explain constant velocity (or speed), making reference to frictional forces.

In space the frictional force is zero.